AP CALCULUS AB

Unit 1 • Lesson 1.6 practice

Lesson 1.6 • Determining Limits Using Algebraic Manipulations

4 multiple-choice questions and 1 free-response question

Question 1

$\lim_{x\to -2} \dfrac{x^2-4}{x+2}$ is

(A) 0

(B) -4

(C) 4

(D) It does not exist.

Question 2

$\lim_{x\to 4} \dfrac{\sqrt{x}-2}{x-4}$ is

(A) $\dfrac{1}{4}$

(B) $\dfrac{1}{2}$

(C) 0

(D) 4

Question 3

$\lim_{h\to 0} \dfrac{(3+h)^2-9}{h}$ is

(A) 9

(B) 0

(C) 6

(D) 3

Question 4

$\lim_{x\to 1} \dfrac{\frac{1}{x}-1}{x-1}$ is

(A) 1

(B) 0

(C) It does not exist.

(D) -1

Question 5 (Free response)

Evaluate each limit. Show the algebra.

(a) $\lim_{x\to 5} \dfrac{x^2-3x-10}{x-5}$

(b) $\lim_{x\to 0} \dfrac{\sqrt{4+x}-2}{x}$

(c) $\lim_{x\to 2} \dfrac{x^3-8}{x-2}$

(d) $\lim_{h\to 0} \dfrac{\frac{1}{2+h}-\frac{1}{2}}{h}$